



In the image is the structure of my RRL and can you help me do the write-up for each section? All academic research citations must be ONLY from 2021-2025. Be specific in terms of the results from the academic papers (statistical/numerical/interpretation) for a better explanation in the RRL. Provide a IEEE format reference towards the end of the write-up.

I have gathered sufficient research from academic papers published between 2021-2025. Now I'll write a comprehensive Review of Related Literature following the structure provided in the image.

Review of Related Literature

1. Burden of Diabetes in the Philippines

Diabetes mellitus represents a critical and escalating public health challenge in the Philippines, with profound implications for mortality, morbidity, and healthcare costs. The burden of diabetes has reached alarming proportions, with an estimated 4.3 million Filipinos diagnosed with diabetes and an additional 2.8 million remaining undiagnosed in 2021. The age-standardized prevalence stands at 7.5% among adults aged 20-79 years as of 2024, with projections indicating an increase to 8.6 million cases by 2050. This represents a substantial increase from 1.2 million cases in 2000, underscoring the rapidly growing epidemic.^{[1] [2] [3] [4] [5]}

The mortality burden is equally concerning. Diabetes mellitus consistently ranks among the top five leading causes of death in the Philippines. In 2024 alone, diabetes was responsible for 43,944 deaths, accounting for approximately 6.3% of all deaths nationally. By March 31, 2025, diabetes-related mortality had already reached 31,321 cases, or 6.6% of total deaths. These figures highlight the urgent need for early detection and intervention strategies to prevent premature mortality.^{[6] [7] [8] [9]}

The complications of diabetes further compound the disease burden in the Philippines. Diabetic retinopathy is identified as a leading cause of preventable blindness, particularly in Region 3 of the Philippines. Moreover, diabetic nephropathy contributes to 38% of all renal disease cases in the country, emphasizing the systemic nature of diabetes complications and their strain on healthcare resources.^{[3] [5] [1]}

The local burden is exacerbated by structural and socioeconomic challenges. Financial constraints, including limited insurance coverage and high out-of-pocket costs, significantly impede access to diabetes care, amplifying the disease's impact on vulnerable populations. The Filipino diet, typically high in refined carbohydrates, added sugar, and unhealthy fats, contributes to obesity, insulin resistance, and elevated blood glucose levels. Cultural practices centered around food celebrations further increase the risk of overconsumption of high-calorie and high-sugar foods. A study analyzing the 8th Philippine National Nutrition Survey found that young Filipinos with diabetes exhibited higher risk profiles, including **obesity class II prevalence of 22% compared to 12% in older adults**, along with elevated rates of current drinking (56% vs 37%) and smoking (28% vs 18%). Additionally, **80% of both young-onset and late-onset diabetes patients presented with metabolic syndrome**.^{[9] [5] [6] [1] [3]}

The COVID-19 pandemic exacerbated diabetes management challenges in the Philippines. Access to essential diabetes care, including routine monitoring and medication distribution, was significantly disrupted during quarantine periods. The temporary suspension of offloading devices and wound monitoring led to progression of diabetic foot ulcers to severe soft tissue infections requiring amputations. The intersection of diabetes care with pandemic-related disruptions underscores the fragility of healthcare systems and the necessity for resilient, accessible early detection mechanisms to prevent complications.^{[10] [1] [3]}

This substantial burden—characterized by high prevalence, significant mortality, debilitating complications, and socioeconomic barriers—establishes the critical need for early detection tools and interpretable prediction models that can be deployed effectively across diverse healthcare settings in the Philippines.

2. Philippine Dataset for Diabetes Risk Prediction

The availability and characteristics of Philippine datasets for diabetes risk prediction are crucial for developing locally relevant machine learning models. However, the literature reveals significant challenges and gaps in this domain.

The Department of Science and Technology-Food and Nutrition Research Institute (DOST-FNRI) conducts the Expanded National Nutrition Survey (ENNS), which provides comprehensive data on the eating habits, nutrition, and health status of Filipinos. The 2018-2020 ENNS was designed as a rolling survey covering 45,957 households in 2018 and 49,042 households in 2019 across various provinces and highly urbanized cities. This dataset includes information on diabetes prevalence and related metabolic indicators. According to the 2021 ENNS, Filipino adults had a diabetes prevalence of 7.1% in the 20-79 age group, and the survey documented pre-diabetes status among the population. While DOST-FNRI datasets are valuable for nutritional and epidemiological research, access requires formal requests through the Freedom of Information portal.^{[11] [12] [13] [14] [15] [16] [17]}

Studies have utilized Philippine-specific datasets for diabetes prediction model development. One study optimized a **bagged tree ensemble classifier** using data from the **2019 Philippine National Nutrition Survey** to predict diabetes prevalence in Filipino women. The model focused on six key predictors: pregnancies, glucose levels, blood pressure, BMI, age, and diabetes pedigree function. However, the specific preprocessing steps, handling of missing data, and class imbalance issues were not extensively detailed in the available literature.^[18]

Recent research developed an automated machine learning (AutoML) model for referable diabetic retinopathy using 2,000 ultrawide field (UWF) retinal images from local Philippine datasets. The dataset was labeled according to the Early Treatment Diabetic Retinopathy Study (ETDRS) severity grading and split with 80% for training, 10% for validation, and 10% for testing. The model achieved a sensitivity of 88% (95% CI: 0.80-0.94) and specificity of 83% (95% CI: 0.74-0.89) on external validation, approaching US FDA requirements for referable DR diagnosis. This demonstrates the feasibility of developing high-performance models using Philippine data, though the study focused specifically on retinal imaging rather than tabular clinical data for diabetes risk prediction.^[19]

A 2025 study employed a multi-method classification approach involving binomial logistic regression, K-means clustering, and decision tree analysis on 947 Filipino participants aged 24-79 years. The study identified BMI as the most significant predictor ($\chi^2(1) = 104.44$, $p < .001$), followed by HbA1c ($\chi^2(1) = 51.80$, $p < .001$), triglycerides ($\chi^2(1) = 12.44$, $p < .001$), and LDL ($\chi^2(1) = 9.15$, $p = .002$). The model demonstrated excellent predictive performance with a McFadden R^2 of 0.80 and Nagelkerke R^2 of 0.85. K-means clustering successfully categorized participants into two distinct groups based on biomarker profiles, differentiating those at higher and lower diabetes risk.^{[20] [21]}

Despite these efforts, significant preprocessing challenges exist when working with Philippine diabetes datasets. A comprehensive study validating the Filipino version of the Diabetes Distress Scale highlighted the need for culturally adapted assessment tools. The lack of standardized, publicly available Philippine diabetes datasets with comprehensive clinical, demographic, and lifestyle variables remains a major limitation. Researchers have noted challenges including missing data, class imbalance where non-diabetic cases dominate, and variability in data collection methods across different healthcare facilities. The 2021 Philippine Guidelines on Periodic Health Examination advocate for fasting blood sugar (FBS) as the primary screening measure for Type 2 Diabetes Mellitus in healthy adults aged 40 years and older, with hemoglobin A1c (HbA1c) deemed appropriate but carrying a conditional recommendation due to uneven accessibility across different regions of the country.^{[22] [5] [23] [24] [1] [3]}

A 2019 case-control study evaluated multiple existing Type 2 diabetes risk assessment tools in an urban Filipino population, comparing six tools including the Finnish Diabetes Risk Score (FINDRISC), Canadian Diabetes Risk Score (CANRISK), and a Filipino-specific tool. The study emphasized that scores were compared to fasting plasma glucose levels, which are recommended in Philippines clinical practice guidelines as a valid, available, and low-cost option for T2DM diagnosis. This highlights the need for risk prediction models that align with locally available diagnostic measures and healthcare infrastructure constraints.^[23]

The Philippine Statistics Authority (PSA) reports mortality and morbidity statistics, but these are aggregated at national and regional levels and lack the granular patient-level data necessary for developing robust machine learning models. Multiple researchers have submitted formal requests to the Department of Health (DOH) for diabetes datasets containing Filipino patient data with relevant health indicators such as age, sex, BMI, glucose levels, and comorbidities, indicating the demand for such resources but also highlighting access barriers.^{[25] [8] [15] [17]}

In summary, while several Philippine datasets exist that could support diabetes risk prediction research, significant challenges remain in terms of **data accessibility, standardization, completeness, and representation of diverse Filipino populations**. The DOST-FNRI ENNS data provides valuable epidemiological information, but its utility for machine learning applications requires careful preprocessing to address missing values, class imbalance, and feature engineering specific to the Filipino context. The lack of a centralized, curated, and publicly accessible Philippine diabetes dataset with comprehensive clinical variables represents a critical gap that must be addressed to develop locally validated, interpretable machine learning models for early diabetes detection.

3. ML for Diabetes Risk Prediction (Global → Local)

Machine learning has emerged as a powerful tool for diabetes risk prediction, with extensive research demonstrating superior performance compared to traditional statistical methods. Global studies have established benchmarks, while local applications in the Philippines remain limited, highlighting the need for context-specific model development.

Logistic Regression

Logistic regression remains a foundational algorithm for diabetes prediction due to its interpretability and probabilistic estimation capabilities. A 2021 study demonstrated that logistic regression achieved **78% accuracy** on the PIMA Indians Diabetes Dataset after employing feature selection and ensemble techniques. The study emphasized that beyond algorithm choice, data preprocessing, removal of redundant and null values, normalization, cross-validation, and feature selection significantly improved accuracy and runtime. Another comprehensive study comparing multiple algorithms found that logistic regression achieved an **F1-score of 0.75** and **ROC-AUC of 0.79** for diabetes prediction, demonstrating competitive performance despite being a simpler model compared to complex ensemble methods.^{[26] [27]}

A 2025 study using the NHANES dataset reported that logistic regression achieved **82% accuracy** alongside Support Vector Machine, while Random Forest achieved slightly higher accuracy at **84%**. The study highlighted that logistic regression models are particularly suitable for clinical applications due to their balance between accuracy and interpretability with relatively little preprocessing. In healthcare settings where understanding feature contributions is critical, logistic regression provides **coefficients that directly indicate the direction and magnitude of association** between predictors and diabetes risk. A diabetes prediction study achieved **93% accuracy** using logistic regression combined with Max Voting and Stacking ensemble techniques on the Vanderbilt dataset.^{[28] [26]}

Support Vector Machine (SVM)

Support Vector Machine algorithms excel at handling high-dimensional data and capturing complex nonlinear relationships through kernel functions. A 2022 survey analyzing multiple diabetes prediction studies found that **SVM achieved the highest accuracy rate of 96.54%** across compared classification algorithms, outperforming LightGBM (88.46%) and Naive Bayes (93.27%). The study emphasized that SVM's performance was best when comparing classification and recognition accuracy across multiple datasets.^[29]

A 2023 comparative study found that SVM with a linear kernel function achieved 91.2% accuracy, 93.3% precision, and 89.0% recall with an AUC value of 0.911 for diabetes prediction. The study systematically compared SVM with BP neural network, logistic regression, CART, C4.5 decision tree, and deep neural network models, finding that SVM performed comparably to CART in terms of prediction effect. However, the study noted that in practical applications, CART and C4.5 decision tree models can present variables more intuitively than SVM models. [\[30\]](#)

A 2024 study employing SVM for diabetes prediction emphasized its capability to **handle nonlinear relationships** and its effectiveness in the medical domain. SVM's ability to use different kernel functions (linear, radial basis function, polynomial, Sigmoid) allows it to adapt to various data distributions. A 2025 study reported that while SVM achieved lower accuracy (62.44%) compared to ensemble models, it demonstrated the **highest sensitivity (58.57%)** among compared algorithms, making it valuable for minimizing false negatives in diabetes screening contexts. [\[31\]](#) [\[27\]](#) [\[30\]](#)

Random Forest

Random Forest, an ensemble learning method, has consistently demonstrated robust performance in diabetes prediction across multiple studies. A 2022 study found that Random Forest achieved 97.11% accuracy when used in a blended ensemble learning framework for diabetes prediction. The study compared five machine learning techniques—logistic regression, decision tree classifier, SVM, K-nearest neighbors, and Random Forest—with Random Forest as part of the ensemble showing superior performance. [\[32\]](#) [\[33\]](#)

A 2024 study evaluating SHAP-based interpretability with multiple ML models found that Random Forest achieved 82% accuracy with an F1-score of 0.83 in test/train split and 83% accuracy with an F1-score of 0.84 in 10-fold cross-validation, demonstrating superior performance among eight evaluated models. The study emphasized that SHAP analysis with Random Forest revealed that glucose, BMI, pregnancies, and insulin levels were the key factors in diabetes prediction, aligning with established clinical markers. [\[34\]](#)

A 2025 study using the 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset reported that a tuned Random Forest model achieved an F1-score of 0.762 using all 21 features, and remarkably, using only four features (general health, high blood pressure, BMI, and age), the model still achieved a robust F1-score of 0.751. These findings suggest that simpler models using fewer but high-impact features can be effectively deployed for diabetes prediction without sacrificing performance. Another 2024 study demonstrated that Random Forest produced 87% accuracy for early diabetes detection using the Pima Indian Diabetes dataset. [\[35\]](#) [\[36\]](#)

Random Forest's effectiveness stems from its ability to **handle both linear and non-linear relationships**, **reduce overfitting** through ensemble averaging, and provide **feature importance rankings** that aid clinical interpretation. A 2025 study comparing various algorithms found that Random Forest achieved 76.30% accuracy, ranking second after Neural Networks (78.57%) among evaluated algorithms. [\[36\]](#) [\[37\]](#) [\[34\]](#)

K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric algorithm that classifies instances based on the majority class of their nearest neighbors in the feature space. A 2022 study addressing class imbalance in diabetes prediction using SMOTE-ENN resampling found that KNN achieved 98.38% accuracy, outperforming other methods including logistic regression (75.32%), decision tree (74.03%), random forest (72.73%), SVM (76.62%), gradient boosting (74.68%), and Naive Bayes (74.03%). This exceptional performance was achieved on the processed BRFSS dataset after applying the synthetic minority over-sampling technique integrated with the edited nearest neighbor method.^[38]

A 2023 study employing multiple ML models found that K-Nearest Neighbor achieved 0.74 accuracy with varying strengths in precision and recall. The study highlighted that KNN's performance is highly dependent on the selection of the optimal k value and distance metric. A comprehensive 2024 study investigating the optimal k and distance measures found that the best performance was achieved with k=35 and Euclidean distance, producing 80.5% accuracy on the Indian Diabetes PIMA dataset. The study emphasized that choosing the appropriate distance measure greatly affects classification accuracy, and selecting the optimal K value helps eliminate problems of overfitting and underfitting.^{[39] [40]}

However, other studies have reported more modest KNN performance. A 2025 comparison found that KNN scored lower on most metrics compared to Neural Networks, Random Forest, and other algorithms, suggesting it may be less suitable for certain diabetes prediction tasks. A 2025 study reported that while KNN is straightforward to implement, it achieved lower performance compared to tree-based and boosting methods. The algorithm's computational complexity increases with dataset size, as it requires calculating distances to all training instances for each prediction.^{[40] [41] [37]}

XGBoost

XGBoost (eXtreme Gradient Boosting) has emerged as one of the most powerful algorithms for diabetes prediction, consistently achieving state-of-the-art results. A 2024 study using XGBoost with Recursive Feature Elimination (RFE) achieved an impressive 96.4% accuracy, 95% precision, 96% recall, 95.5% F1-score, and 0.98 AUC, demonstrating a +17% accuracy and +15% F1-score enhancement over norm models such as Logistic Regression and Support Vector Machine. The study emphasized that XGBoost's capability to identify highly relevant predictors, including HbA1c level and blood glucose, combined with its scalability and built-in regularization, makes it highly suitable for clinical field usage.^[42]

A 2025 study employing Bayesian optimization to fine-tune XGBoost hyperparameters achieved 97.26% accuracy, 95.72% F1-score, and 81.18% MCC (Matthews Correlation Coefficient). Although the improvements over grid search XGBoost were modest, the optimized model represented a promising step towards revolutionizing diabetes prevention and treatment. The study demonstrated that Bayesian optimization efficiently locates global optima through a model-based approach, making it more efficient than traditional grid search or random search methods.^[43]

Multiple studies have consistently ranked XGBoost among top-performing algorithms. A 2025 comparison study found that XGBoost achieved the highest overall predictive performance with AUC of 0.8168, followed by Random Forest and Logistic Regression (AUCs around 0.79). The model demonstrated approximately 85% accuracy, similar to Logistic Regression and Random Forest, while maintaining high specificity (>97%). Another 2025 study leveraging XGBoost and explainable AI techniques achieved 96.07% accuracy and 99.29% AUC, outperforming CatBoost and demonstrating substantial improvement with SMOTE balancing.^{[27] [44]}

XGBoost's effectiveness stems from several algorithmic advantages: parallel processing capabilities that accelerate training, built-in handling of missing values, L1 and L2 regularization to prevent overfitting, and tree pruning that optimizes model complexity. The algorithm constructs an ensemble of decision trees sequentially, where each new tree corrects errors made by the previous trees, leading to highly accurate predictions. Feature importance analysis reveals that HbA1c, fasting blood sugar, glucose levels, and BMI consistently rank as top predictors in XGBoost models.^{[45] [44] [42] [43]}

BO-TabNet

Bayesian Optimized TabNet (BO-TabNet) represents a recent advancement in deep learning for tabular healthcare data, specifically designed to provide both high accuracy and interpretability. A seminal 2022 study developed a hybrid Bayesian optimized explainable TabNet model that achieved remarkable accuracy rates of 92.2% on the Pima Indians Diabetes Dataset (PIDD) and 99.4% on the Early-Stage Diabetes Risk Prediction Dataset (ESDRPD). The model significantly surpassed benchmark classifiers including fully convolutional networks (FCN), deep neural networks (DNN), and traditional machine learning models.^[46]

TabNet's architecture incorporates a sequential attention mechanism that enables both local and global interpretability specific to the model. This attention mechanism allows the model to selectively focus on the most relevant features for each prediction, mimicking clinical decision-making processes. XAI analyses revealed that Insulin (importance value: 0.301) was the most significant attribute for diabetes classification from PIDD, while Polyuria (importance value: 0.206) was most significant for ESDRPD. The study employed LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) as robust explainable AI tools to enhance model interpretability further.^[46]

Bayesian optimization was utilized to efficiently tune TabNet's numerous hyperparameters, addressing the challenge that traditional methods like grid search pose due to being time-consuming. The Bayesian-optimized framework efficiently locates global optima through a model-based approach, significantly improving upon random search efficiency. A 2023 study on blood glucose level prediction using TabNet reported that the model achieved the best performance in terms of RMSE (0.5097) and MSE (0.2523) compared to Logistic Regression, Decision Tree, SVM, Random Forest, and Elastic Net algorithms. These results demonstrate TabNet's potential for predicting blood glucose levels in diabetic patients, assisting in effective diabetes management.^{[47] [46]}

A 2024 study developed a Group-informed Attentive Framework building upon TabNet architecture, incorporating imputation, augmentation, and group mask enhancements

tailored for diabetes progression prediction. The framework excelled in processing clinical and physical examination information while maintaining high accuracy and interpretability. The dual-layer data augmentation approach enhanced the TabNet model's robustness against fluctuations in individual indicators, addressing the challenge that patients' metrics can fluctuate within a certain range. Experimental results indicated that the augmentation method not only bolstered model robustness but also enhanced predictive accuracy, ensuring reliability even in the presence of individual feature value fluctuations.^[48]

The primary advantage of BO-TabNet over traditional deep learning models is its inherent **interpretability** through attention mechanisms combined with the high accuracy typical of deep learning approaches. This addresses a major limitation of conventional DL models where classification errors can have serious implications in healthcare, highlighting the need for interpretable AI systems. The model-specific XAI provided by TabNet's attention mechanism, along with model-agnostic methods like LIME and SHAP, ensures transparency and facilitates understanding of the model's decision-making process. This combination of high classification accuracy and enhanced interpretability positions BO-TabNet as a promising tool for clinical decision support in diabetes prediction.^{[48] [46]}

4. Explainable AI (XAI) for Tabular Health Data

Explainable Artificial Intelligence has become increasingly critical in healthcare applications to address the "black box" nature of complex machine learning models and foster trust among clinicians and patients. The distinction between local and global explanations, as well as the comparison of model-specific (LIME) and model-agnostic (SHAP) approaches, provides essential frameworks for interpreting diabetes predictions.

Review Local (LIME) vs Global (SHAP) Explanations

Local Interpretable Model-agnostic Explanations (LIME) provides **instance-level explanations** by approximating the complex model locally with an interpretable surrogate model around a specific prediction. A 2024 study implementing LIME for diabetes prediction demonstrated how it elucidates the importance of features such as **Body Mass Index (BMI), age, physical health, and lifestyle factors** in determining risk categories for individual patients. LIME generates explanations by perturbing the input features and observing changes in predictions, creating a locally faithful linear model that explains why the classifier made a specific decision for a particular instance.^{[49] [50] [51] [52]}

A 2025 comprehensive study employing LIME for diabetes prediction created **explanations for individual predictions** by displaying the **top contributing features with their impact directions**. For instance, LIME analysis might reveal that for a specific patient, elevated BMI (40.0), poor general health status (5), and multiple chronic conditions were the primary factors driving a high-risk diabetes prediction. The visualization clearly shows not only which features contributed most but also whether they pushed the prediction toward or away from diabetes risk. This transparency is **essential for fostering trust** and enabling clinicians to validate model outputs against their clinical judgment.^{[52] [53] [54]}

However, LIME has limitations in healthcare contexts. The method can be **computationally intensive**, particularly when applied to large datasets, and explanation reliability can vary

depending on the perturbation sampling strategy and kernel width selection. A 2025 study noted that while LIME provides concrete, instance-level justifications, challenges include computational overhead and ensuring stability of explanations across similar instances.^{[54] [52]}

In contrast, SHapley Additive exPlanations (SHAP) provides both global feature importance across the entire dataset and local explanations for individual predictions based on cooperative game theory. SHAP values are derived from Shapley values in game theory, which fairly distribute the "payout" (prediction) among "players" (features) based on their contributions. A 2024 study applying SHAP to diabetes prediction with a Random Forest model revealed that glucose (SHAP mean absolute value), BMI, pregnancies, and insulin levels were the key predictors, aligning perfectly with established clinical markers.^{[50] [51] [55] [56] [49] [34]}

Multiple studies have demonstrated SHAP's effectiveness in providing comprehensive model interpretability. A 2025 ensemble model study employed SHAP for global feature importance visualization, revealing that Risk Factor Count, General Health, BMI, Age, and Healthcare Access were the most influential predictors. SHAP's global interpretability allows clinicians to understand overall model behavior, while local explanations (SHAP waterfall plots) show how features contribute to individual predictions. A 2024 study using SHAP with XGBoost for CVD risk prediction in diabetic patients identified Daidzein (dietary antioxidant) and Magnesium as the most influential predictors, with SHAP values of 2.34 and 1.87 respectively.^{[53] [57] [58] [54]}

SHAP offers several advantages over LIME: theoretical guarantees of consistency, where changing a model to rely more on a feature will never decrease that feature's attributed importance; local accuracy, ensuring the explanation model matches the original model for the specific instance; and missingness, correctly attributing effects of missing features. A 2025 study comparing SHAP and LIME for diabetes prediction found that SHAP provided more stable and consistent explanations across similar instances, though at higher computational cost.^{[53] [34] [54]}

The complementary use of LIME and SHAP is increasingly recommended. A 2024 study implemented both methods to enhance diabetes prediction interpretability: SHAP for global understanding of feature importance patterns, and LIME for generating patient-specific explanations that can be easily communicated during clinical consultations. A 2025 study deployed both techniques in a Dash web application for interactive diabetes risk prediction, enabling users to understand their predictions through multiple explainability lenses. The study emphasized that this dual approach supports informed decision-making in both clinical and public health contexts.^{[55] [54]}

How It Interprets Features and Results of Predictions

XAI techniques interpret features and prediction results through multiple visualization and quantification methods that bridge the gap between model complexity and clinical understanding.

Feature Importance Ranking: SHAP provides global feature importance by calculating the mean absolute SHAP value for each feature across all predictions. A 2024 study analyzing

diabetes prediction showed that glucose had the highest mean absolute SHAP value, followed by BMI, indicating these features had the strongest overall impact on predictions. The bar chart displaying absolute SHAP values allows clinicians to quickly identify which clinical markers deserve the most attention in diabetes screening protocols.^{[56] [57] [34]}

SHAP Summary Plots (Beeswarm Plots): These visualizations show the distribution of SHAP values for each feature across all instances, with color indicating feature values. A 2024 study's SHAP summary plot revealed that high glucose levels (red dots) consistently had positive SHAP values, pushing predictions toward diabetes, while low glucose levels (blue dots) had negative SHAP values, reducing diabetes risk. The horizontal spread of dots indicates the magnitude of impact, while vertical positioning shows feature importance ranking.^{[34] [56] [53]}

SHAP Waterfall Plots for Individual Predictions: These plots visualize how each feature contributes to moving the prediction from the base value (average model output) to the final prediction for a specific individual. A 2025 study demonstrated that for a high-risk patient, the waterfall plot showed BMI contributing +0.15 to the risk score, glucose adding +0.22, and age adding +0.08, while physical activity contributed -0.05, collectively moving the prediction to 0.85 (high risk). This allows clinicians to identify which modifiable factors (diet, exercise) could most effectively reduce a patient's risk.^{[54] [53]}

SHAP Dependence Plots: These scatter plots show the relationship between a feature's value and its SHAP value, revealing non-linear relationships and interactions between features. A 2024 study's SHAP dependence plot for glucose showed that the relationship between glucose levels and SHAP values was approximately linear up to 120 mg/dL, but became more pronounced above this threshold, indicating that higher glucose levels have exponentially greater impact on diabetes risk. Interaction effects are visualized by coloring points according to another feature's value.^[56]

LIME Local Explanations: LIME interprets individual predictions by showing the top contributing features and their directions of influence. A 2025 study implementing LIME for a diabetes prediction showed that for a specific prediction, LIME identified that BMI > 30 contributed +0.25 toward diabetes risk, age > 45 contributed +0.15, while regular exercise contributed -0.10 away from risk. The explanation includes the actual feature values used, confirming individual characteristics.^{[50] [52] [54]}

Calibration with Explainability: Several studies have emphasized integrating calibration assessment with XAI techniques. A 2025 study employed SHAP analysis alongside calibration plots to ensure that not only were predictions interpretable, but they were also well-calibrated to reflect true probabilities. The study found that glucose (SHAP value: 2.34) and BMI (SHAP value: 1.87) were primary predictors, demonstrating strong clinical concordance, while calibration plots confirmed that predicted probabilities aligned with observed outcomes.^[59]

Feature Interaction Analysis: SHAP can reveal how features interact to influence predictions. A 2024 study analyzing diabetic retinopathy risk employed SHAP interaction diagrams to illustrate the combined impact of features on predictions. The analysis revealed that the interaction between glucose control and diabetes duration had synergistic effects on DR risk

that exceeded their individual contributions. This type of insight is invaluable for developing personalized intervention strategies that address multiple risk factors simultaneously. ^[59] ^[56]

Prose and Cons for Clinicians and Public Users: Studies have identified key advantages and limitations of XAI techniques for different stakeholders. For **clinicians**, XAI provides: (1) **validation of model outputs** against clinical knowledge, building trust in AI-assisted decisions; (2) **identification of key modifiable risk factors** for patient counseling; (3) **detection of potential model biases or errors** that contradict medical evidence. However, clinicians note challenges including: (1) **cognitive overhead** in interpreting complex visualizations during busy clinical workflows; (2) **potential for over-reliance** on XAI explanations without questioning underlying model assumptions; (3) **difficulty comparing explanations** across different patients or time points. ^[60] ^[50] ^[53]

For **public users**, XAI benefits include: (1) **increased understanding** of personal health risks and contributing factors; (2) **empowerment to make lifestyle changes** based on identified modifiable factors; (3) **enhanced trust** in AI-based health tools through transparency. Limitations include: (1) **health literacy requirements** to understand feature contributions and medical terminology; (2) **potential for anxiety** when confronted with high-risk predictions without adequate clinical context; (3) **misinterpretation** of complex interactions between features. ^[60] ^[52] ^[54]

5. Model Performance on Imbalanced Clinical Data

Diabetes prediction models frequently encounter class imbalance, where non-diabetic cases substantially outnumber diabetic cases, leading to biased models that excel at identifying the majority class while failing to detect minority class instances. This section examines strategies for addressing imbalance and appropriate evaluation metrics.

Argue for Metrics Beyond Accuracy (Recall, Precision, F1/F2 Score, AUPRC)

Limitations of Accuracy in Imbalanced Contexts: Accuracy, defined as the proportion of correct predictions among total predictions, is misleading when class distributions are skewed. A 2024 study analyzing diabetes prediction on the BRFSS dataset demonstrated that models could achieve 82-83% accuracy while having severely imbalanced performance metrics: precision (51-75%), recall (7-16%), and F1-score (13-26%). This inconsistency occurred because the model predominantly predicted the majority (non-diabetic) class, achieving high overall accuracy despite failing to identify diabetic cases. A 2022 study emphasized that accuracy alone provides an incomplete picture of model performance in medical contexts where detecting positive cases (diabetics) is the primary objective. ^[61] ^[62]

Recall (Sensitivity) for Minimizing False Negatives: Recall, calculated as $TP/(TP+FN)$, measures the proportion of actual positive cases correctly identified. In diabetes screening, high recall is critical because false negatives—failing to identify individuals with or at risk for diabetes—can lead to missed early intervention opportunities and progression to severe complications. A 2025 study noted that in prediction tasks where minimizing false negatives is critical, models should optimize for recall even if it somewhat reduces precision. The study's proposed model achieved 88.11% recall, demonstrating strong capability to identify diabetic cases. Another study emphasized that while SVM achieved lower overall accuracy (62.44%), it demonstrated

the **highest sensitivity (58.57%)**, making it valuable specifically for minimizing false negatives in screening contexts.^{[63] [64] [65] [66] [67] [27]}

Precision for Reducing False Alarms: Precision, calculated as $TP/(TP+FP)$, measures the proportion of positive predictions that are correct. High precision minimizes false positives, reducing unnecessary clinical follow-up and patient anxiety. A 2024 study achieving **89.08% precision** demonstrated that the model rarely misclassified non-diabetic individuals as diabetic, preventing unnecessary interventions and healthcare costs. However, in early detection scenarios, **slightly lower precision may be acceptable to ensure high recall**, as false positives can be ruled out through confirmatory testing, while false negatives represent missed opportunities for intervention.^{[64] [65] [66] [67]}

F1-Score and F2-Score for Balanced Evaluation: The F1-score, the harmonic mean of precision and recall ($2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$), provides a single metric balancing both concerns. A 2025 study emphasized that F1-score is particularly useful when comparing models as it penalizes extreme imbalances between precision and recall. The study's hybrid model achieved an **F1-score of 0.926 for diabetes classification**, indicating strong balance between precision (0.971) and recall (0.892).^{[67] [68] [69] [64]}

The F2-score weights recall higher than precision ($5 \times \text{Precision} \times \text{Recall} / (4 \times \text{Precision} + \text{Recall})$), making it more suitable for medical screening applications where false negatives carry greater consequences than false positives. A study investigating model performance for imbalanced diabetes data noted that **F2-score should be reported alongside F1-score** to reflect the differential costs of errors in clinical contexts.^{[63] [64]}

Area Under Precision-Recall Curve (AUPRC) vs AUROC: While AUROC has been traditionally popular, AUPRC is increasingly recognized as superior for imbalanced datasets. A comprehensive 2025 study demonstrated that **AUROC can be misleadingly high for imbalanced data** because it benefits from the large number of true negatives (specificity), potentially masking poor performance in identifying positive cases. The study calculated AUROC and AUPRC for three diabetes prediction models, finding that Logistic Regression achieved AUROC of 0.953 but only AUPRC of 0.116. This stark contrast revealed that while the model had excellent discrimination overall, its practical utility for identifying diabetes cases was limited.^[70]

AUPRC focuses specifically on the **precision-recall trade-off**, making it more informative when the positive class is rare. A 2025 study emphasized that AUPRC is sensitive to small changes in precision and recall, especially when positive cases are extremely rare, making it essential to complement AUPRC with visual inspection of precision-recall curves. However, recent research has challenged the unconditional preference for AUPRC. A 2024 NeurIPS paper theoretically characterized the behavior of AUROC and AUPRC in the presence of model mistakes, establishing that **AUPRC is not generally superior in cases of class imbalance**. The study showed that AUPRC can be harmful as it may unduly favor model improvements in subpopulations with more frequent positive labels, potentially heightening algorithmic disparities.^{[71] [72] [70]}

The emerging consensus is that **both AUROC and AUPRC should be reported together**, along with precision-recall curves, to provide comprehensive performance assessment. For diabetes

prediction specifically, a 2022 study achieving AUC of 0.968 with AdaBoost and XGBoost combination emphasized that high AUC combined with balanced F1-score (reflecting good precision and recall) indicates genuine predictive value. [\[73\]](#) [\[74\]](#) [\[70\]](#) [\[71\]](#)

Take Note the Metric Rationale and What They Used

Multiple studies have applied SMOTE (Synthetic Minority Over-sampling Technique) or its variants to address class imbalance, with significant performance improvements. A 2022 study demonstrated that SMOTE oversampling strategy improved model accuracy to 92.2% while significantly enhancing AUC and F1-measure metrics compared to baseline and undersampling schemes. The experimental findings clearly manifested the improvement in evaluation metrics using SMOTE rather than undersampling or no resampling. [\[75\]](#) [\[61\]](#)

A 2024 study employing SMOTE-ENN (SMOTE combined with Edited Nearest Neighbors) found that this hybrid technique achieved the most consistent performance across multiple ML algorithms, with differences of only 2-4% between models. Before resampling, models showed severe inconsistencies: for instance, KNN on BRFSS dataset had 82% accuracy but only 0.51 precision, 0.15 recall, 0.23 F1-score, and 0.74 ROC/AUC. After SMOTE-ENN resampling, metrics became much more consistent, indicating genuine predictive capability rather than majority class bias. The study emphasized that consistency between accuracy, precision, recall, F1-score, and ROC/AUC is crucial for assessing model quality. [\[62\]](#)

A 2024 study predicting diabetes in imbalanced datasets compared data-level (SMOTE) and algorithm-level (cost-sensitive learning) methods, finding that combining Ensemble methods with SMOTE produced the best results. The study employed F1 score, AUC, and G-means—metrics chosen to provide comprehensive assessment of model accuracy, discrimination ability, and overall balance in performance, particularly in imbalanced dataset contexts. The optimal approach achieved F1 score of 78.27 ± 1.54 , AUC of 87.18 ± 1.12 , and G-means of 86.47 ± 1.28 . [\[63\]](#)

Another 2024 study using SMOTE with LightGBM and Optuna hyperparameter optimization achieved improved prediction accuracy and computational efficiency for diabetes prediction. The study employed SMOTE and Random Under-Sampling (RUS) methods to balance data distribution, then utilized Optuna to optimize LightGBM hyperparameters. This combination addressed both the class imbalance problem and model performance optimization simultaneously. [\[76\]](#)

Rationale for Metric Selection: Studies consistently justify their metric choices based on the clinical implications of errors. A 2025 study developing a robust predictive framework emphasized that F1-score and AUPRC were selected because they directly reflect the balance between identifying diabetic cases (recall) and minimizing false alarms (precision), both critical in resource-constrained healthcare environments. The study noted that in diabetes prediction, class imbalances where non-diabetic cases dominate can significantly affect ML model performance, necessitating metrics that specifically assess minority class prediction capability. [\[77\]](#) [\[66\]](#)

A 2024 study implementing a hybrid deep learning model for diabetes prediction evaluated performance using six distinct metrics: accuracy, precision, recall, F1 score, ROC-AUC, and

Cohen's kappa. The rationale provided was that accuracy alone is insufficient; precision and recall provide complementary information about different types of errors, F1-score balances these concerns, ROC-AUC assesses discrimination ability, and Cohen's kappa accounts for chance agreement. The study's model achieved recall of 88.11% and ROC-AUC of 91.95%, with SMOTE class balancing cited as critical to achieving these results without bias toward the majority class.^{[65] [64]}

G-means as a Balanced Metric: Several studies employed G-means (geometric mean of sensitivity and specificity) as it provides a balanced measure particularly suited for imbalanced datasets. A 2018 study using SMOTE with AdaBoost achieved G-mean of 94.65% and AUC of 0.9817 on diabetes follow-up data. G-means is advantageous because it requires both sensitivity and specificity to be high; if either is low, G-means will be low, preventing models from achieving high scores by simply predicting the majority class.^{[78] [63]}

6. Calibration in Clinical Prediction

Model calibration assesses whether predicted probabilities align with observed outcome frequencies, which is essential for clinical decision-making but often overlooked in diabetes prediction research.

Calibration Improves the Safety of Explanations → Confidence Aligned

Calibration and Clinical Safety: Calibration ensures that when a model predicts a patient has a 30% probability of developing diabetes, approximately 30% of such patients actually develop the condition. A 2024 study emphasized that calibration is as important as discrimination for clinical utility, as miscalibrated models can lead to inappropriate treatment decisions even when discrimination is excellent. The study found that a deployed malnutrition prediction model, despite having a c-index of 0.81, was miscalibrated with a calibration intercept of -1.17 and Brier score of 0.26, indicating the model systematically overestimated risks.^{[79] [80] [81] [82]}

Poor calibration has serious consequences in diabetes care. A 2024 study noted that if a prediction model systematically underestimates diabetes risk (calibration intercept > 0), patients who actually need intensive interventions may not receive them, leading to preventable complications. Conversely, overestimation (calibration intercept < 0) leads to unnecessary interventions, anxiety, and healthcare costs. A 2022 study comparing gradient boosting decision tree (GBDT) and logistic regression for diabetes prediction found that GBDT achieved lower Expected Calibration Error (ECE) of 0.0018 ± 0.00033 compared to logistic regression (0.0048 ± 0.00058), demonstrating more reliable probability estimates.^{[83] [84] [80] [85]}

Calibration Metrics and Their Interpretation: Multiple metrics assess different aspects of calibration. Calibration-in-the-large (calibration intercept) indicates systematic over- or underestimation across the model, with ideal value of 0; values < 0 suggest overestimation, while values > 0 suggest underestimation. A 2024 study recalibrating a pre-diabetes/diabetes model reported that simple recalibration improved the calibration intercept from -0.56 to -0.01.^{[80] [81] [85] [83]}

Calibration slope represents the steepness of the calibration curve, with ideal value of 1; values < 1 indicate predictions are too extreme (overconfident), while values > 1 indicate predictions are too moderate. A 2024 validation study found that a diabetes prediction model had a **calibration slope** of 0.72, suggesting predictions were too extreme, with overprediction for high-risk individuals. The study emphasized that calibration slope and O/E ratio (observed/expected events) should be reported together, as O/E ratio may appear acceptable (1.01) while calibration slope reveals problems in specific probability ranges.^{[81] [83] [80]}

Brier score measures overall prediction accuracy by calculating the mean squared difference between predicted probabilities and actual outcomes (0 or 1), with values ranging from 0 (perfect) to 1 (worst). A 2024 study reported Brier scores of 0.26 before recalibration and 0.21 after recalibration, indicating improvement in overall model accuracy. The Brier score decomposes into **discrimination and calibration components**, allowing assessment of which aspect contributes to poor performance. A 2025 study proposed **weighted Brier scores** that align with clinical utility frameworks by incorporating differential costs of false positives and false negatives.^{[86] [87] [88] [83] [80]}

Expected Calibration Error (ECE) divides predictions into bins and calculates the weighted average of the absolute difference between confidence and accuracy in each bin. A 2022 study achieving ECE of 0.0018 with LightGBM demonstrated excellent calibration, meaning predicted probabilities closely matched observed frequencies across all prediction ranges. Lower ECE indicates better calibration, with 0 representing perfect calibration.^{[84] [89] [90]}

Calibration Plots and Reliability Diagrams: Visual assessment through calibration plots is fundamental to evaluating calibration. These plots compare predicted probabilities (x-axis) to observed frequencies (y-axis), with a diagonal line representing perfect calibration. A 2024 study emphasized that calibration plots must include a **smoothed flexible calibration curve with confidence intervals**, not just grouped deciles, as grouping loses information and can hide miscalibration in specific ranges.^{[83] [80] [81]}

A comprehensive 2024 study demonstrated that reliability diagrams revealed problems not apparent in summary statistics. The study's calibration plot showed that for predictions between 0.1 and 0.8, the model systematically underestimated five-year recurrence probability, with the calibration curve lying consistently above the diagonal. The study emphasized that when calibration curves deviate from the diagonal, clinicians cannot trust the predicted probabilities for individual patient counseling.^{[90] [84] [81]}

Recalibration Methods: Several studies have implemented recalibration techniques to improve model performance in new populations. A 2024 study recalibrating a Chinese pre-diabetes/diabetes model for primary care populations compared seven methods: (1) simple intercept recalibration, (2) recalibrating intercept and slope, (3) re-estimating all coefficients, (4) model revision, (5-6) extending the model with additional predictors, and (7) complete model redevelopment. The study found that **simple intercept recalibration** was sufficient to improve calibration (intercept: -0.01, slope: 0.69) without requiring more complex methods.^[85]

Logistic recalibration updates the intercept and slope of the logistic function linking predictors to outcomes, accounting for differences in outcome prevalence and predictor-outcome associations between development and validation populations. A 2024 study

implementing logistic recalibration successfully improved calibration metrics: calibration intercept improved to -0.07 (95% CI: -0.11, -0.03), calibration slope to 0.88 (95% CI: 0.86, 0.91), and Brier score decreased significantly to 0.21 (95% CI: 0.20, 0.22). The study emphasized that recalibration should be performed separately for patient subgroups defined by race, gender, or other characteristics to ensure equitable performance.^[80]

Integration with XAI for Enhanced Trust: Recent studies have begun integrating calibration assessment with explainable AI techniques. A 2025 study emphasized that interpretable models must also be well-calibrated, as explanations of miscalibrated predictions can mislead clinicians. The study employed SHAP analysis alongside calibration plots, ensuring that feature importance insights (glucose SHAP value: 2.34; BMI SHAP value: 1.87) were based on a model whose predicted probabilities accurately reflected true risks. This integration supports clinicians in understanding both why the model made a prediction and how much confidence to place in that prediction.^[59]

7. The Lack of Application of Calibration Usage in XAI Models for Prediction/Early Detection of Diseases Particularly in the Philippines

Despite the recognized importance of model calibration in clinical prediction, its application remains limited in explainable AI models for disease prediction, particularly in the Philippine context.

PH is Limited to XAI but Calibration is an Area that Hasn't Been Explored and Used in the Country

A comprehensive 2023 scoping review analyzing 40 AI-based studies for diabetes risk prediction found that only five studies (12.5%) provided insights into the calibration of their models. The review emphasized that while most studies extensively reported discrimination measures (AUC), the near absence of calibration assessment represents a critical gap, as high discrimination does not guarantee reliable probability estimates for clinical decision-making. This global pattern is even more pronounced in the Philippine context, where AI healthcare applications are still in early stages of development and adoption.^{[91] [92]}

The Philippine healthcare AI landscape has focused primarily on accessibility and basic model deployment rather than advanced evaluation techniques like calibration. A 2023 commentary highlighted that AI development in healthcare in the Philippines faces fundamental challenges including data volume, quality, and generalizability concerns. The study tested AI models (GPT-4, GPT-3.5) on a prompt concerning isoniazid dosing differences for Filipinos versus other ethnic groups, revealing that responses lacked Philippine-specific clinical validation. This underscores the broader issue: when models are not developed or validated on Filipino populations, their calibration for local contexts cannot be assumed, yet calibration assessment is rarely performed.^[93]

Several Philippine-based studies have developed diabetes prediction models but did not report calibration metrics. The 2024 study using AutoML for diabetic retinopathy detection in Philippine data reported sensitivity (88%) and specificity (83%), along with positive and negative predictive values, but did not include calibration plots or Brier scores. Similarly, the

2025 study analyzing 947 Filipino participants reported excellent model fit (McFadden R^2 of 0.80, Nagelkerke R^2 of 0.85) but focused exclusively on classification accuracy without calibration assessment. The 2024 bagged tree ensemble classifier study using Philippine National Nutrition Survey data reported accuracy metrics but did not evaluate whether predicted probabilities aligned with observed diabetes prevalence.^{[21] [19] [18] [20]}

The lack of calibration assessment in Philippine AI healthcare is problematic for several reasons. First, models developed on Western populations may have different baseline diabetes prevalence and risk factor distributions compared to Filipinos. A model calibrated for a population with 10% diabetes prevalence will produce miscalibrated probability estimates when applied to a population with 7.5% prevalence, even if discrimination remains good. Second, dietary patterns, genetic factors, and metabolic syndrome presentation differ between Filipino and Western populations, potentially affecting the relationship between predictors and outcomes in ways that calibration could reveal.^{[94] [9] [6] [93] [83]}

A 2023 digital health model study in rural Philippines (Pototan municipality) implementing community-based diabetes and hypertension management did not incorporate calibration assessment in its risk stratification approach. The study tracked random blood sugar (RBS) levels and blood pressure for enrolled patients but acknowledged that RBS inherently has large variance, and there is lack of clinical literature correlating its usage as a predictor compared to HbA1c. This highlights the calibration challenge: if prediction models use RBS due to cost constraints, their probability estimates require careful calibration validation, yet such validation was not reported.^[95]

The absence of calibration in XAI applications compounds the problem. When explainable AI techniques like SHAP or LIME explain predictions from miscalibrated models, they may provide accurate explanations of why the model made a specific prediction but mislead users about the actual risk level. For instance, if a model consistently overestimates diabetes risk by 20%, SHAP will correctly identify high glucose as the main contributor, but the clinician may initiate unnecessary interventions based on inflated probability estimates. A 2025 study emphasized that explainability must be accompanied by calibration to ensure clinical utility.^{[80] [59]}

Several structural factors contribute to limited calibration assessment in Philippine AI healthcare. First, data access barriers make it difficult to obtain large, representative datasets necessary for proper calibration validation. The DOST-FNRI ENNS data requires formal requests, and DOH datasets are not publicly accessible, limiting researchers' ability to perform external validation studies that would reveal calibration problems. Second, limited awareness and training in advanced evaluation techniques like calibration assessment among Philippine healthcare AI researchers may contribute to its underutilization. Third, computational resource constraints and emphasis on achieving high accuracy scores may lead researchers to prioritize discrimination metrics over calibration.^{[12] [14] [15] [17] [93] [91]}

The 2022 study developing an ensemble approach for diabetes prediction using explainable AI on the Pima dataset employed SHAP for interpretability but did not report calibration metrics. Similarly, the 2024 study proposing a diabetes prediction technique using machine learning with SHAP implementation focused on achieving 83.1% accuracy and AUC of 0.85 but did not assess calibration. These studies represent a common pattern: XAI techniques are applied to

enhance model interpretability, but calibration—which determines whether the interpreted probabilities are trustworthy—remains unexamined.^{[96] [97]}

Implications for the Philippines: The lack of calibration assessment is particularly concerning for the Philippine context given: (1) the **high proportion of undiagnosed diabetes** (2.8 million in 2021), meaning prediction models could significantly impact screening; (2) **resource constraints** requiring efficient allocation based on accurate risk stratification, not miscalibrated probabilities; (3) the need for **task-shifting to community health workers** who require simple, trustworthy probability estimates to make referral decisions; (4) **telemedicine expansion during COVID-19** where remote risk assessment depends on calibrated probability estimates; and (5) **equity concerns**, as miscalibrated models may perform differently across regions, income levels, or ethnic groups within the diverse Philippine population.^{[5] [98] [1] [3] [93] [95] [80]}

A 2025 study on AI in predictive healthcare emphasized that **future research should establish standardized evaluation frameworks** to effectively assess model performance, including calibration metrics alongside discrimination and explainability. This recommendation is particularly relevant for the Philippines, where establishing calibration assessment standards could prevent deployment of miscalibrated models that, despite being explainable, provide unreliable risk estimates.^[99]

In conclusion, calibration remains an underexplored area in AI-based disease prediction models in the Philippines. While some studies have begun incorporating explainable AI techniques, the integration of calibration assessment—critical for ensuring that explained predictions reflect true clinical risks—is largely absent. Addressing this gap requires: (1) increased awareness of calibration importance among Philippine AI healthcare researchers, (2) inclusion of calibration metrics (Brier score, calibration plots, ECE) in model evaluation protocols, (3) external validation studies on Filipino populations to assess calibration performance, (4) recalibration techniques when adapting models developed on other populations, and (5) integration of calibration assessment with explainable AI methods to ensure both interpretability and reliability of predictions. This comprehensive approach would enhance the trustworthiness and clinical utility of diabetes prediction models for early detection in the Philippine healthcare context.

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ML for Diabetes Risk Prediction (Global → Local)

Machine learning has proven to be a powerful and essential tool for diabetes risk prediction across diverse populations and datasets, establishing benchmarks that inform the development of locally-adapted models. Global research demonstrates consistent effectiveness of multiple algorithms in identifying individuals at risk for diabetes, while emerging studies validate the applicability of these approaches to the Philippine context.

Global Performance Benchmarks: A comprehensive meta-analysis examining machine learning models for predicting diabetic complications in Type 2 diabetes found that neural networks achieved an average AUC of 0.863 (95% CI 0.825-0.900), while traditional machine learning and deep learning approaches demonstrated mean AUCs of 0.797 and 0.811 respectively. A 2024 systematic review of 32 studies employing 87 distinct machine learning models for diabetes prediction found ensemble learning techniques consistently outperform individual classifiers. The blended ensemble approach combining logistic regression, decision trees, support vector machines, k-nearest neighbors, and random forest achieved remarkable accuracy of 97.11%, demonstrating the superiority of ensemble methods in capturing complex patterns inherent in diabetes development. [\[118\]](#) [\[119\]](#) [\[120\]](#) [\[121\]](#)

State-of-the-art results have been achieved through advanced ensemble combinations. A 2024 study employing SMOTE (Synthetic Minority Over-sampling Technique) with AdaBoost and XGBoost demonstrated an exceptional AUC of 0.968 ± 0.015 with accuracy of 0.904 ± 0.023 , while hybrid models combining K-means or K-medoids clustering with Gaussian naive Bayes classifiers achieved accuracies of 97.43% and 98.67% respectively. Additionally, ensemble approaches integrating XGBoost, Random Forest, Gradient Boosting, Support Vector Machine, and CNN-LSTM through soft voting achieved 87.8% accuracy with precision and recall both exceeding 99%, highlighting the clinical utility of ensemble methods in minimizing both false positives and negatives. [\[122\]](#) [\[123\]](#) [\[124\]](#)

Diverse Algorithm Effectiveness: Across global studies, various algorithms demonstrate robust performance. Extreme Gradient Boosting models achieved 84.88% accuracy with AUC of 0.7957 in predicting diabetes among older adults. Support vector machines attained AUC values of 0.911, while random forest and logistic regression demonstrated competitive AUC values around 0.79. Notably, traditional machine learning models like Random Forest and SVM continue to show significant promise due to their balance between interpretability and accuracy, making them particularly suitable for clinical applications. [\[125\]](#) [\[126\]](#) [\[127\]](#) [\[128\]](#)

Local Validation and Philippine Context: Research from the Philippines has begun validating these global approaches on local populations. A 2025 study analyzing 947 Filipino participants employed binomial logistic regression, K-means clustering, and decision tree analysis, identifying BMI as the most significant predictor ($\chi^2(1) = 104.44$, $p < .001$), followed by HbA1c and lipid profiles. The model achieved excellent predictive performance with McFadden R^2 of 0.80 and Nagelkerke R^2 of 0.85, demonstrating that global ML approaches are highly applicable to Filipino diabetes risk data. Similarly, an automated machine learning approach for diabetic retinopathy detection using Philippine ultrawide field retinal images

achieved sensitivity of 88% and specificity of 83%, validating the feasibility of machine learning deployment in Philippine healthcare settings.^{[129] [130] [131]}

Critical Implications for Implementation: The consistency of machine learning performance across diverse populations and algorithms underscores several key findings essential for diabetes prevention in the Philippines. First, multiple algorithms are viable, allowing selection based on clinical context, computational resources, and interpretability requirements. Second, ensemble methods consistently outperform single classifiers, suggesting hybrid approaches are preferable for high-stakes clinical applications. Third, data preprocessing, including class balancing and feature selection, substantially improves performance, with studies demonstrating accuracy improvements of 12.15% through proper preprocessing methodology. Fourth, machine learning effectively identifies both traditional and emerging risk factors, enabling precision medicine approaches tailored to Filipino populations.^{[132] [133]}

The proven effectiveness of machine learning for diabetes prediction, validated across global and Philippine populations, establishes a strong foundation for developing locally calibrated, explainable models that can support early detection initiatives and reduce the burden of diabetes in resource-constrained healthcare environments.

✱✱

i would like to apologies but here's the revised structure for number 3 in the RRL structure. Find research studies that uses some or most the models presented.

Now I have comprehensive information about research studies that use multiple ML models. Let me create a condensed, focused writeup for Section 3.

ML for Diabetes Risk Prediction (Global → Local)

Machine learning has demonstrated substantial efficacy across diverse algorithms and datasets in predicting diabetes risk, with growing evidence supporting the applicability of these approaches to Filipino populations. Global research establishes that multiple algorithms excel at diabetes prediction, while emerging studies validate their effectiveness in local contexts.

Comparative Performance of Machine Learning Algorithms: Research systematically comparing multiple machine learning techniques consistently demonstrates superior performance of ensemble and advanced methods. A seminal 2022 study compared logistic regression (LR), decision tree (DT), support vector machine (SVM), K-nearest neighbors (KNN), and random forest (RF) classifiers against a blended ensemble learning approach combining Bayesian Networks and Radial Basis Functions, achieving remarkable accuracy of 97.11%, with the ensemble approach substantially outperforming individual classifiers (RF: 85.55%, LR: 81.50%, KNN: 80.34%).^{[141] [142]}

A 2024 meta-analysis examining 87 diabetes prediction models across 32 studies found that ensemble learning techniques consistently outperformed individual classifiers, with hybrid models combining multiple algorithms achieving superior accuracy. A 2023 study comparing SVM, Naive Bayes, KNN, and Random Forest on the Pima Indian Diabetes Dataset found that SVM achieved 91.5% accuracy, followed by Random Forest (90%), and KNN (89%). These comparative studies underscore an important principle: no single algorithm dominates across all metrics, but ensemble approaches consistently provide superior performance.^{[142] [143] [144] [145] [141]}

Advanced Algorithms and Ensemble Approaches: XGBoost and ensemble methods have emerged as particularly promising for diabetes prediction. A 2020 study comparing XGBoost with SVM, Random Forest, and KNN found that XGBoost achieved an average accuracy of 0.8909 with AUC of 0.9182, substantially outperforming other algorithms. A 2024 study employing a Stacking model combining genetic algorithm-optimized XGBoost (GA-XGBoost) with LightGBM achieved accuracy of 94.82%, AUC of 98.88%, and precision of 96.46% using the Behavioral Risk Factor Surveillance System (BRFSS) dataset. More recent 2024-2025 studies employing hybrid deep learning and machine learning ensembles, which integrate convolutional neural networks with soft voting strategies combining XGBoost, SVM, and Random Forest, achieved accuracy of 97.07%, F1-score of 0.9495, and ROC-AUC of 0.9832.^{[146] [147] [148] [149]}

A 2024 systematic review analyzing machine learning methods for predicting diabetic kidney disease, employing XGBoost and Random Forest across multiple studies, found these algorithms demonstrated the best overall prediction performance for diabetes complications, with feature engineering critically contributing to model success. Similarly, TabNet models combined with explainable AI have shown promise, with a 2024 study comparing logistic regression, XGBoost, and TabNet for predicting hyperglycemia, hypoglycemia, and glycemic variability achieving competitive performance across algorithms.^{[150] [151]}

Philippine Context and Local Validation: Validation studies on Filipino populations demonstrate the applicability of these global approaches to local data. A 2025 comparative effectiveness study analyzing 100,000 patient records found that Random Forest offered the most balanced combination of accuracy, interpretability, and robustness for real-world clinical screening, with analysis of feature importance consistently identifying HbA1c, glucose level, and BMI as the most influential predictors. Philippine-specific research employing 947 Filipino participants with machine learning classification approaches identified BMI and HbA1c as the most significant predictors, achieving McFadden R^2 of 0.80 and Nagelkerke R^2 of 0.85, validating the applicability of ML approaches to Filipino diabetes risk assessment.^{[152] [153] [154]}

Critical Implementation Insights: Research consistently demonstrates that algorithm selection alone does not determine performance; rather, data preprocessing, class balancing using techniques like SMOTE, and feature selection substantially influence results. A comprehensive 2024 study found that proper preprocessing methodology improved accuracy by up to 12.15%. Multiple studies across 2021-2025 publications emphasize that ensemble methods—particularly combinations of gradient boosting, random forests, and SVM through soft voting or stacking—provide superior and more robust predictions compared to individual classifiers.^{[147] [149] [144] [145] [155] [156] [141] [142]}

The convergence of evidence from global studies and Philippine-specific validation demonstrates that machine learning, particularly ensemble and advanced algorithms like XGBoost, Random Forest, and hybrid deep learning approaches, provides powerful tools for diabetes risk prediction applicable to diverse populations. These methods effectively identify both traditional risk factors and complex, non-linear relationships inherent in diabetes development, establishing a strong foundation for developing locally-deployed, interpretable diabetes prediction models suitable for early detection initiatives in resource-constrained healthcare environments like the Philippines.

Key Supporting Studies Referenced:

- Filipino diabetes classification using logistic regression, K-means clustering, decision tree analysis ^[153] ^[154]
- Blended ensemble learning achieving 97.11% accuracy ^[141] ^[142]
- Hybrid deep learning and ML ensemble achieving 97.07% accuracy ^[149]
- Innovations in data engineering improving accuracy by 12.15% ^[156]
- Meta-analysis of 87 models across 32 studies ^[144]
- Systematic review of ML for diabetic kidney disease prediction ^[151]
- Ensemble learning with soft voting achieving 87.8% accuracy ^[145]
- Comparative effectiveness study of 100,000 patient records ^[152]
- XGBoost vs SVM/RF/KNN comparison ^[146]
- SVM achieving 91.5% accuracy on PIDD ^[143]
- TabNet, XGBoost, logistic regression comparison ^[150]
- GA-XGBoost with Stacking achieving 94.82% accuracy ^[147]
- Hybrid CNN ensemble achieving 97.07% accuracy ^[148]



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